

(Data Source Requirements)

Created: 2026-03-08 **Purpose:** Technical requirements for each external data source needed by the P20 MCP server **Feeds into:** P20 MCP (Phase 2: Structured Data)

Data Source Priority Ranking

Priority	Data Source	Questions Dependent	P20 Tool Mapping
1	ERP —	5 questions	query_inventory
2	ERP — /	4 questions	query_orders, query_sales
3	ERP — /	5 questions	query_costs, query_financials
4	CRM — /	6 questions	query_customer, query_supplier
5	ERP —	4 questions	query_production
6	External — /	3 questions	query_market_prices
7	ERP —	4 questions	query_procurement
8	ERP —	1 question	query_contracts

DS-1: ERP System

System Information

- **System name:** [To be confirmed — likely SAP, Kingdee, or UFIDA]
- **Deployment:** [On-premise / Cloud]
- **Modules needed:** Inventory, Sales, Finance/Cost, Production, Procurement, Contracts
- **Access method:** [API / Database query / Report export]

Required Data Points

Inventory Module

Data Point	Used By	Access Pattern
Current stock by material/product	Q-002, Q-033	Real-time query
Safety stock levels	Q-033	Configuration lookup
Stock aging / days of supply	Q-033, Q-037	Calculated field
WIP (work in progress)	Q-035	Real-time query
Inventory turnover by product	Q-037	Calculated over period

Sales/Order Module

Data Point	Used By	Access Pattern
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Order history by customer	Q-011	Historical query
Order pipeline / backlog	Q-018, Q-035	Real-time query
Sales by rep / product / period	Q-016	Aggregate query
Accounts receivable by customer	Q-013	Real-time query
Delivery schedule	Q-018, Q-035	Real-time query

Cost/Finance Module

Data Point	Used By	Access Pattern
Product cost breakdown (material, labor, energy, depreciation)	Q-039	Period query
Product margins	Q-004, Q-040	Calculated
Budget vs actual by department	Q-010, Q-043	Period query
Cash flow summary	Q-041	Period query
AR/AP days outstanding	Q-041	Calculated

Production Module

Data Point	Used By	Access Pattern
Production order status	Q-018	Real-time query
Yield rate by line / period	Q-019, Q-025	Aggregate query
Equipment downtime log	Q-020	Historical query
Capacity utilization by line	Q-021	Calculated
Material consumption vs standard	Q-022	Variance query
Production schedule	Q-024, Q-035	Real-time query

Procurement Module

Data Point	Used By	Access Pattern
Purchase history by item/person/dept	Q-001	Historical query
Purchase price history by material	Q-006	Time series query
Supplier on-time delivery rate	Q-009	Aggregate query
Contract expiry dates	Q-007	Lookup query

Access Requirements

- **Read-only access:** All MCP queries are read-only. No write operations.
- **Authentication:** Service account with minimum necessary permissions
- **Rate limiting:** Expect ~100 queries/day during normal use, bursts during evaluation

- **Data freshness:** Ideally real-time; daily sync acceptable for initial phase
- **Security:** Data must not leave the corporate network; MCP runs locally

DS-2: CRM System

System Information

- **System name:** [To be confirmed]
- **Deployment:** [On-premise / Cloud / WeChat-based]

Required Data Points

Data Point	Used By	Access Pattern
Customer profile (name, industry, size, credit terms)	Q-004, Q-014	Lookup
Supplier profile (rating, history, capabilities)	Q-003, Q-005	Lookup
Cooperation history (orders, performance scores)	Q-003	Historical query
Customer complaints log	Q-015	Historical query
Sales targets by rep	Q-016	Period query

Access Requirements

- Same as ERP: read-only, service account, local network

DS-3: External Market Data

Data Sources

Source	Data Point	Used By	Update Frequency
Shanghai Futures Exchange	Copper futures prices	Q-002, Q-006	Daily
LME	Aluminum prices	Q-006	Daily
Industry price index	PET film, target material prices	Q-006	Weekly/Monthly
Company credit databases (/)	Supplier/customer credit info	Q-003, Q-014	On-demand

Access Requirements

- **API access:** Free tier may suffice for daily price lookups
- **Cost:** Budget for commercial API if needed (API ~¥5000/year)
- **Caching:** Cache daily prices locally to reduce API calls

DS-4: Future Data Sources (Deferred)

Source	Data Point	Questions	Blocker
Email system	Supplier correspondence history	Q-003	Privacy, access policy
Meeting records	Meeting notes, decisions	Q-003	Format, digitization
Lab/experiment DB	Test results, experiment parameters	Q-023, Q-029	System existence unclear
HR system	Headcount, recruitment pipeline	Q-047	Privacy, access policy
Customs/logistics	Shipment tracking, clearance status	Q-038	Third-party system
Patent database	Patent filings and claims	Q-030	External API cost

Action Items

1. **Confirm ERP system type and version** — needed to determine API approach
2. **Confirm CRM system type** — or whether customer data is in ERP
3. **Request read-only API/database access** to ERP for P20 MCP server
4. **Identify IT contact** who can provide API documentation and test credentials
5. **Evaluate external market data APIs** — test free tiers, estimate costs for paid tiers